Perplexity-Guided Sampling for Self-Evolving Curriculum in RL-based LLM Fine-Tuning

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Introduction

Motivation:

- RL fine-tuning significantly enhances LLM reasoning (e.g., OpenAI-o1, DeepSeek-R1).
- Training curriculum (data order & selection) is a critical bottleneck.
- Existing dynamic curricula (e.g., SEC) use Multi-Armed Bandit for cluster selection but sample **randomly** inside clusters.
- **Our finding**: Combining SEC with intra-cluster PPL sorting hurts performance due to rigid cluster boundaries.
- **Question**: Can we design a simpler, direct sample-level curriculum?

Self-Evolving Curriculum (SEC)

Formulates curriculum learning as a Non-stationary Multi- Armed Bandit (MAB) problem, adapting sampling distribu- tion based on real-time learning progress.

- Reward Signal: Batch-average of Absolute Advantage (via GRPO).

$$r_t(c) = \mathbb{E}\left[|\hat{A}_t|\right] \tag{1}$$

- Selection: Boltzmann Sampling based on Q-values.

$$p(c) \propto e^{Q_t(c)/\tau} \tag{2}$$

- Update: Q-values are updated via Temporal Difference (TD(0)).

$$Q_{t+1}(c) = \alpha \cdot r_t(c) + (1 - \alpha) \cdot Q_t(c) \tag{3}$$

Bottle neck: Selects the right category (difficulty level), but defaults to Uniform Random Sampling within the cluster which ignores fine-grained sample difficulty.

Uncertainty Driven Curriculum

Previous studies:

- High entropy in reasoning corresponds to greater sample diversity, making model training more robust.
- Collapse of Policy Entropy: The sudden collapse of policy entropy induces over-confidence, causing the model to converge prematurely and fail to achieve superior performance.
- Most existing studies use Entropy as a proxy of uncertainty, which may be biased to longer responses.

Why Perplexity?

- Entropy:

$$H(y|x) = -\sum_{t=1}^{L} \log p_\theta(y_t|y_{<t}, x) \tag{4}$$

- Perplexity:

$$\text{PPL}(y|x) = \exp\left(-\frac{1}{L}\sum_{t=1}^{L} \log p_\theta(y_t|y_{<t}, x)\right) \tag{5}$$

- Perpleixty length-normalized $\rightarrow$ avoids bias toward long responses.

Efficient estimation:

- QAP: modifies the input prompt to instruct the model to answer concisely within a few steps. Prompt: *Answer concisely within three reasoning steps.*
- Prefix Approximation:

$$\text{PPL} \approx \exp\left(-\frac{1}{L}\sum_{t=1}^{L} \log p_{\theta_k}(y_t|y_{<t}, p_{\text{quick}}, x)\right) \tag{6}$$

Perplexity-Guided Sampling (PGS)

We observed that uncertainty-driven sampling within SEC yields underwhelming performance. Consequently, we propose the lightweight alternative PGS.

1. Global ranking: Sort entire dataset by perplexity (PPL) descending.
2. Batch construction: Take top-B highest-PPL samples at each training step.
3. Periodic PPL update (every I steps) using: Quick-Answer Prompting (QAP) for efficiency & Prefix PPL approximation (length-normalized)

Key idea: Directly challenge the model on its most uncertain samples.

Algorithm 1 Perplexity-Guided Sampling (PGS)

Input: Training set $\mathcal{D}$ partitioned into categories $C = \{c_1, ..., c_N\}$, LLM policy π_θ, Learning rate α, Temperature τ, Batch size B, Total steps T, Prefix length L, Update interval I
Output: Fine-tuned LLM π_θ
Initialize PPL$[x_i] \leftarrow 0$ for all samples in $\mathcal{D}$
for $t \leftarrow 0$ **to** $T-1$ **do**
  $\mathcal{B}_t \leftarrow \emptyset$
  if $t \bmod I = 0$ **then**
    for each sample x_i in $\mathcal{D}$ **do**
      $x_i' \leftarrow \text{QAP}(x_i)$
      Generate prefix $\tilde{y}_i \sim \pi_\theta(\cdot|x_i')$
      PPL$[x_i] \leftarrow$ Prefix PPL Approximation (Eq. 5)
    end for
    Sort $\mathcal{D}$ by PPL$[x_i]$ in descending order
  end if
  Select top B samples from $\mathcal{D}$ to form $\mathcal{B}_t$
  Run π_θ on each $x \in \mathcal{B}_t$ to generate rollouts and compute rewards r
  Estimate advantages $\hat{A}$ and update π_θ with RL algorithm
end for
Return: Fine-tuned LLM π_θ

Experimental Setup

Base model: Qwen2.5-Math-1.5B

Task: Countdown (use numbers + $\{+, -, \times, \div\}$ to reach target)

Dataset: 30k samples, 3 difficulty levels - Easy (3 numbers), Medium (4 numbers), Hard (5 numbers) – 10k each

Test sets:
In-distribution (ID): 200 each of Easy/Medium/Hard
Out-of-distribution (OOD): Very Hard (6 numbers, 200 samples)

Baselines: Random, SEC Bandit, SEC + Intra-cluster PPL

RL algorithm: GRPO, 140 steps, batch size 32

PGS settings: Update interval I=10, sample size 10k (33% of data), prefix length 128

Reward design:

Score	Condition
0.0	Format error
0.1	Correct format, but numbers don't match
0.2	Format error, but numbers match
0.3	Correct format & numbers, but wrong answer
1.0	Correct format, numbers, and answer

Main Results

Table: Main results on Countdown test sets. OOD refers to Very Hard (6 numbers).

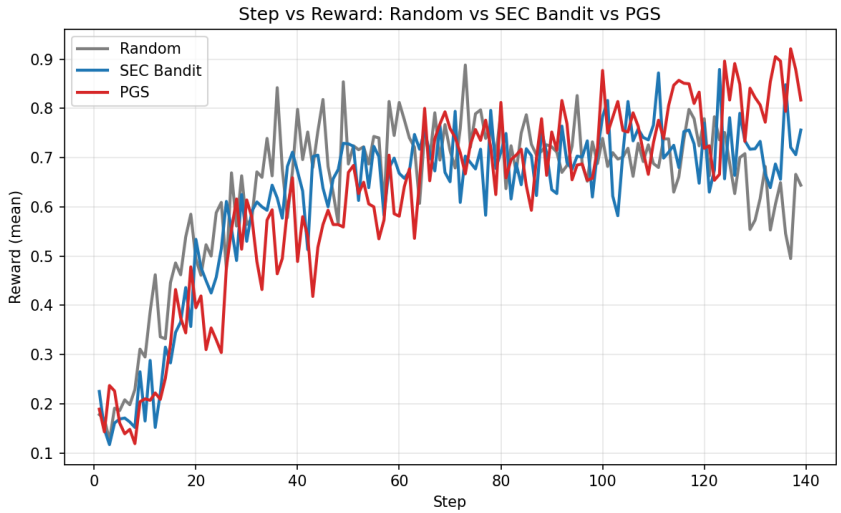
Method	Easy (ID)	Medium (ID)	Hard (ID)	Very Hard (OOD)
Random	79.5	56.2	46.7	34.1
SEC Bandit	90.9	67.5	61.9	44.8
SEC + Intra-cluster PPL	88.0	66.1	59.8	38.2
PGS (Ours)	**95.1**	**70.5**	**67.8**	**48.5**

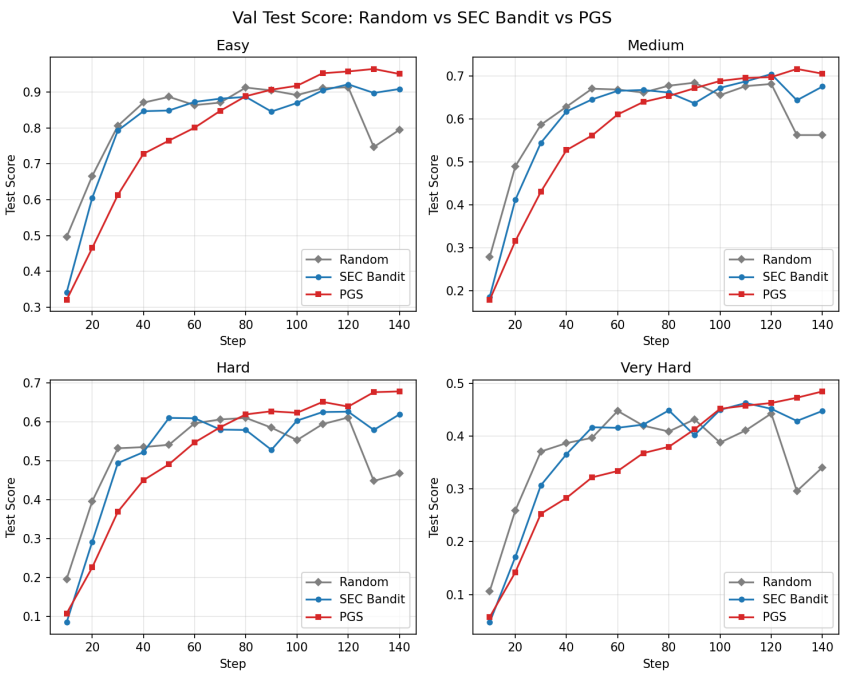
- PGS outperforms all baselines on every ID and OOD test.
- Absolute gain over SEC: +4.2% (Hard ID), +3.7% (OOD).
- SEC + Intra PPL underperforms even random on OOD $\rightarrow$ shows harmful interaction.

Analysis: Main Result

Two-phase learning:

- Early phase: PGS lags behind SEC and Random → Weak policy overwhelmed by high-uncertainty samples.
- Late phase: PGS catches up and surpasses all baselines → Persistent training on uncertainty frontier prevents plateauing.
- Short-term sacrifice for higher final performance.





Analysis: Hybrid Case

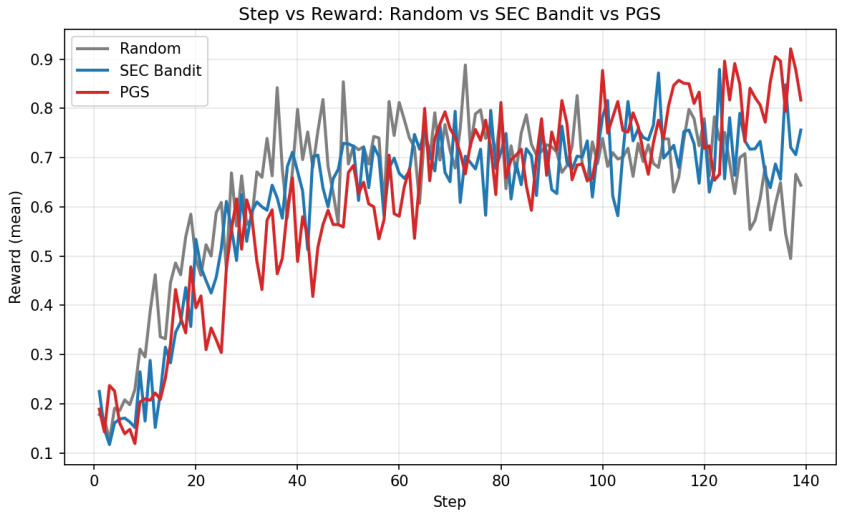
Rigid cluster confinement:

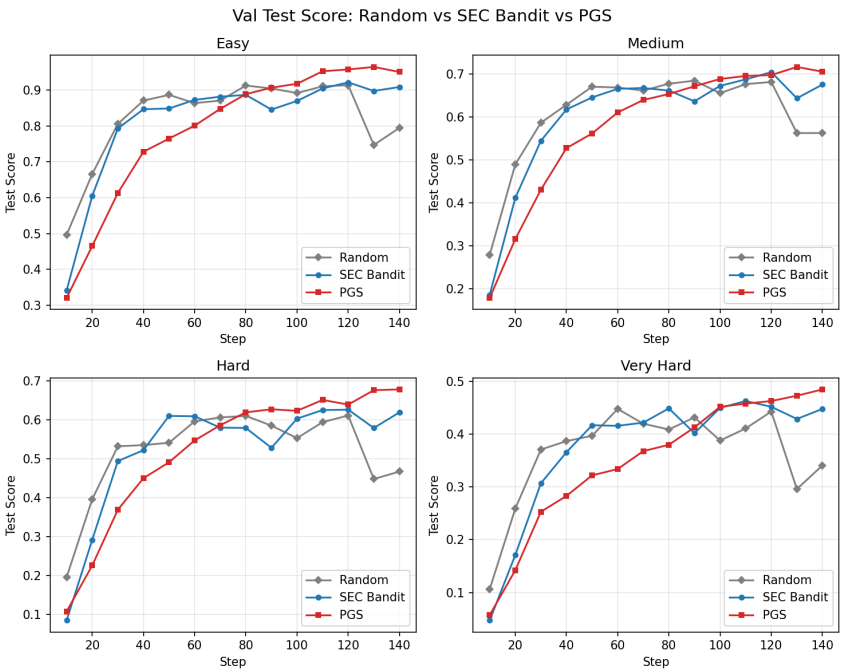
- Model is forced to stay inside the selected cluster.
- Cannot access hard samples from other clusters even if they are more informative.

Early-stage uncertainty overload without escape:

- Hybrid method feeds high-PPL samples from a narrow domain.
- Unlike pure PGS, no cross-cluster diversity $\rightarrow$ learning stalls.

Result: Lower final performance than vanilla SEC and PGS.





Limitations

- **Narrow task & model scope**: Experiments only on Countdown dataset and only on Qwen2.5-Math-1.5B model.
- **Small modeling ceiling**: Small models possess inherently weaker reasoning capabilities, may constrain the evaluation of method.
- **Future work**: Broader benchmarks (GSM8K, MATH), larger models (7B, 14B), diversed scheduling.

Conclusion

We propose PGS – a lightweight, sample-level curriculum based on global perplexity ranking, removing the bandit scheduler entirely.

- PGS consistently outperforms Random, SEC, and hybrid variants on Countdown (ID & OOD).
- Combining SEC with intra-cluster PPL is harmful due to rigid boundaries.
- PGS exhibits a two-phase dynamic: initial lag, then superior convergence.

Thanks!

Questions?

Q&A